

MARKETCAPTURE

End-to-End System Flow

How one market-data packet becomes a deterministic order book and durable evidence

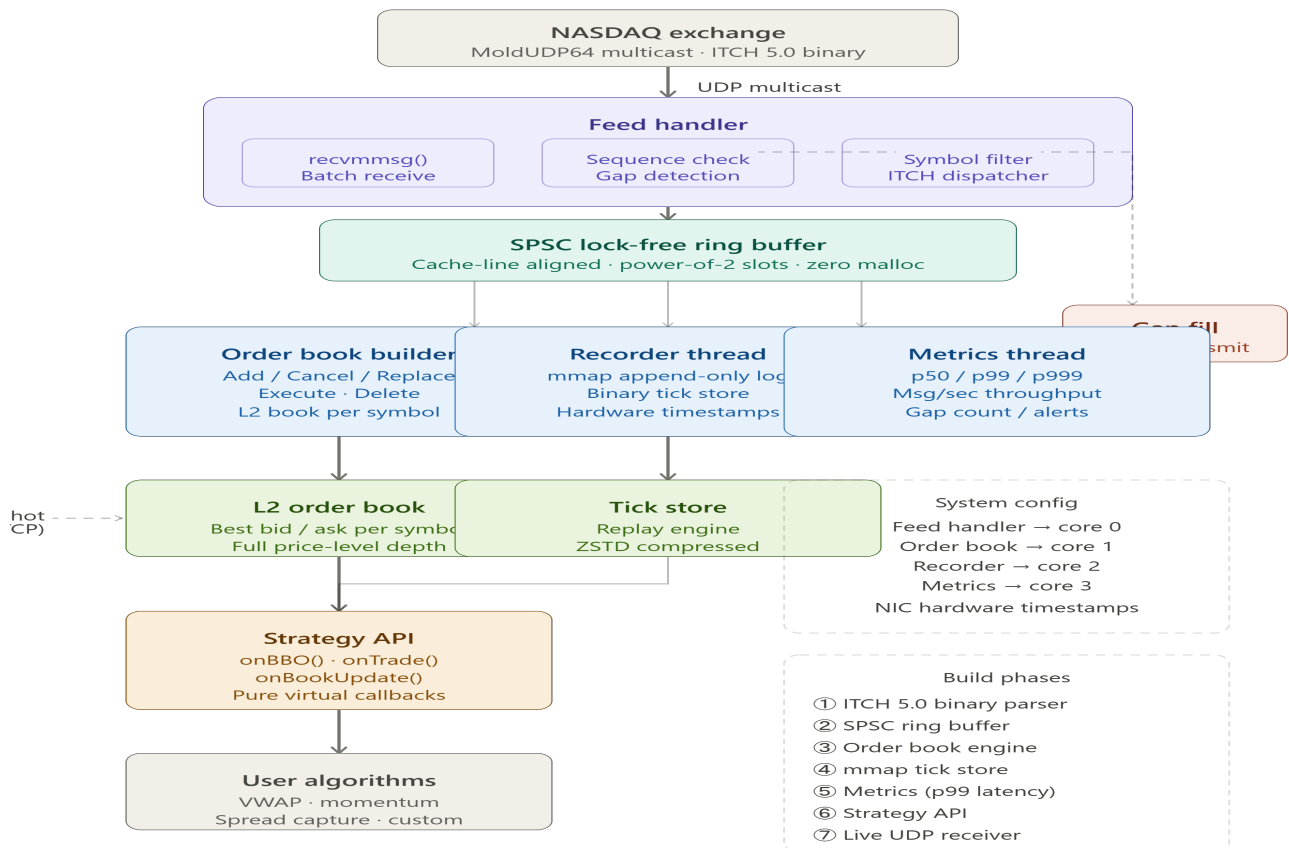
A visual operational guide for developers, reviewers, and interviewers

C++20 | NASDAQ ITCH 5.0 | MoldUDP64 | deterministic recovery | bounded-memory hot path

Reading map

Stage	Purpose	Thread / ownership
NIC receive	Accept UDP/DPDK/io_uring input	Single RX producer
Arbitration	Merge A/B, suppress duplicates, block gaps	Parser worker
ITCH parsing	Validate binary messages and normalize events	Parser worker
Fan-out	Publish bounded values to independent consumers	SPSC queues
Book	Maintain order and price-level state	Arena-book worker
Recorder / PCAP	Persist normalized events and raw packets	Dedicated workers
Metrics	Count progress, errors, rejects, and queue pressure	Metrics worker

System architecture



The original architecture is extended in v2 with fixed-width hot events, bounded arbitration, arena-backed books, asynchronous PCAP, and explicit overload telemetry.

1. Receive and timestamp

Hot-path rule: no filesystem calls, console output, compression, or book mutation occurs in the NIC callback.

The live receiver joins a provider-assigned multicast group or accepts frames through DPDK. The submitter performs bounded work: optional timestamp acquisition, one copy into a preallocated packet slot, and publication of the slot index.

A rejected submit means the fixed pool is exhausted or shutdown has started. The reject policy protects the RX thread; the spin policy applies backpressure when the deployment explicitly chooses it.

Packet lifecycle

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NIC -> fixed packet slot -> ingress SPSC -> parser consumes -> slot returned to free pool
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2. Decode MoldUDP64

The decoder validates the 20-byte MoldUDP64 header, reads session, first sequence, and message count, then creates non-owning spans for length-prefixed ITCH messages.

Malformed lengths, truncation, and trailing bytes are rejected. Spans remain valid only while the packet slot is owned by the parser stage.

3. A/B arbitration and recovery

Channel A and B carry redundant sequence streams. A bounded circular reorder window stores fixed 64-byte ITCH payloads. A duplicate sequence is dropped. A future sequence blocks delivery and raises a missing-range request.

Recovered packets re-enter through the recovery channel. Delivery resumes only in exact sequence order, which preserves deterministic downstream state.

Decision sequence

Condition	Action
sequence < expected	Count duplicate; discard
sequence == expected	Deliver and drain consecutive buffered messages
sequence > expected	Buffer; request [expected, sequence-1]
window collision / capacity	Fail explicitly; never overwrite silently

4. Parse ITCH into HotEvent

HotItchParser validates message type and exact length. Order-impacting messages become a fixed-width HotEvent containing numeric fields plus 8-byte symbol and 4-byte attribution arrays.

Administrative and auction messages are validated but represented as informational on the order-building hot path. The full typed parser remains available for control-plane consumers.

Order state transitions

ITCH	Book effect
A / F	Create order and increase price level
E / C	Reduce shares; delete fully executed order
X	Cancel shares
D	Delete remaining order
U	Atomically replace old reference with new reference/price/size

5. SPSC fan-out

Queues are bounded. Bounded systems expose overload; they do not hide it behind unbounded memory growth.

The parser is the single producer for book, recorder, and metrics queues. Each queue has one dedicated consumer. Acquire/release publication makes the event visible without mutexes.

High-watermarks show the maximum observed occupancy. They are evidence for capacity planning and reveal which consumer is limiting throughput.

6. Book, recorder, metrics, and PCAP

The arena-book worker is the only owner of mutable book state. Hash nodes come from a recycling PMR pool backed by a fixed arena with no general-heap fallback.

The recorder worker materializes legacy string-bearing events only after leaving the hot path. The PCAP worker writes raw packet copies asynchronously. The metrics worker updates progress independently.

7. Checkpoint, restart, and replay

CheckpointStore writes two alternating generations with checksums, file flush, atomic replacement, and directory durability. On restart it selects the newest valid generation and falls back if corruption is detected.

PCAP and normalized tick replay preserve recorded order. The deterministic simulator and recovery source reproduce packet loss, retransmission, restart, and final book equivalence.

8. Shutdown sequence

Shutdown stops new ingress, drains the parser, marks parser completion, drains every consumer, flushes and closes recorders, joins threads, and publishes final book counts.

This ordering prevents a successful shutdown from silently abandoning accepted work.

Operational success criteria

- packets_submitted equals the intended accepted input count
- messages_parsed, book_updates, and metrics_events agree for one-message simulator packets
- rejected packets, parse errors, and unrecovered gaps are zero for a clean run
- checkpoint restart produces identical symbol and active-order counts
- queue watermarks remain below sustained saturation with production headroom

Evidence boundary

The repository proves software behavior in CI. Licensed-feed compatibility, NIC line rate, PTP timestamp accuracy, and FPGA DMA performance require the actual provider network and hardware.